

Stored Procedures (Self-Learning)

Q. What is a Stored Procedure? Explain how to create and execute them, discuss parameters, and highlight the advantages of using stored procedures in a database.

📌 **Definition to Remember**

A **Stored Procedure** is a pre-compiled, named block of SQL statements and procedural logic (IF/ELSE, loops) stored directly on the database server. Applications invoke it by name using **EXECUTE** or **CALL**, rather than sending raw SQL over the network. Used in Oracle (PL/SQL), SQL Server (T-SQL), and MySQL.

1. Creating and Executing a Stored Procedure

Syntax:

```
CREATE PROCEDURE Procedure_Name (Parameters...)
AS
BEGIN
    -- SQL statements and procedural logic
END;
```

Execution:

```
EXECUTE Procedure_Name;    -- SQL Server
CALL Procedure_Name();     -- MySQL / PostgreSQL
```

2. Parameters in Procedures

Parameter Type	Direction	Description
IN	Caller → Procedure	Passes a value into the procedure (default)
OUT	Procedure → Caller	Returns a value back to the caller
INOUT	Both directions	Accepts and modifies a value, then returns it

Example:

```
CREATE PROCEDURE Give_Raise (IN emp_id INT, IN raise_amount DECIMAL)
AS
BEGIN
    UPDATE Employee SET Salary = Salary + raise_amount WHERE ID = emp_id;
END;

-- Call the procedure
CALL Give_Raise(101, 5000);
```

3. Advantages of Stored Procedures

Advantage	Explanation
Pre-compilation	Compiled once; subsequent calls use the pre-compiled plan → faster
Reduced Network Traffic	Client sends one procedure call instead of many SQL lines
Security	Users can EXECUTE a procedure without needing direct SELECT/UPDATE on tables → prevents SQL Injection
Code Reusability	Business logic is centralized; change the procedure once, all apps use updated logic

★ Must-Write Points (for 10 marks)

1. Stored Procedure = named, pre-compiled block of SQL + procedural logic stored in the database.
2. Created with **CREATE PROCEDURE** ; executed with **EXECUTE** or **CALL** .
3. Three parameter types: **IN** (input), **OUT** (output), **INOUT** (both).
4. **Pre-compilation** makes procedures faster than raw SQL queries for repeated operations.
5. **Reduced Network Traffic**: one call replaces hundreds of SQL lines sent over the network.
6. **Security**: users can execute a procedure without direct table access, preventing SQL Injection.
7. **Code Reusability**: centralized business logic — update once, all applications benefit.

⚡ Quick Recall

Stored Procedure → Pre-compiled → CREATE/EXECUTE → Parameters (IN/OUT/INOUT) → Advantages: Faster + Reduced Network Traffic + Security + Code Reusability